

The Problems of Underfitting and Overfitting

Chapter 4: Training Models

Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, Gurelien Geron, 3e, O'Reilly Media, 2022.

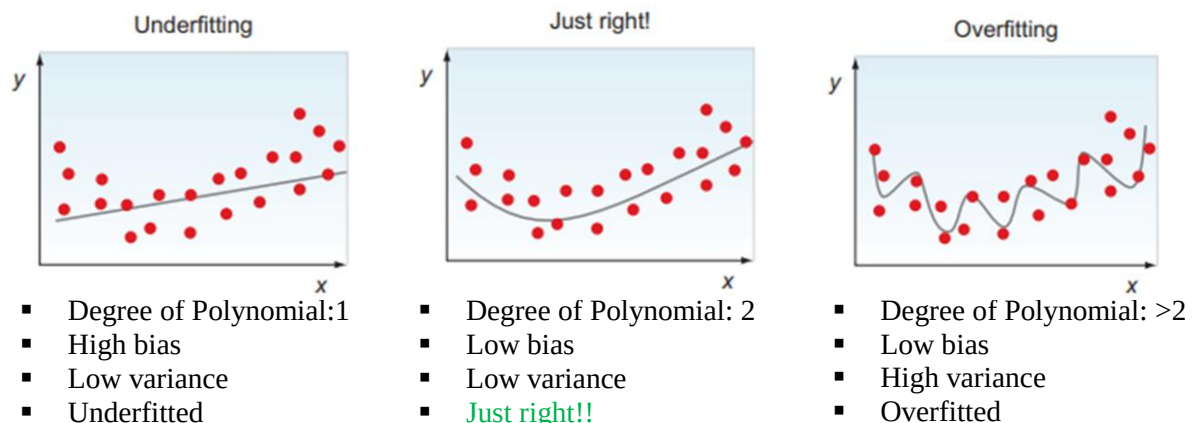
Generalization Error and the Bias-Variance Tradeoff

$$\text{generalization error} = \text{bias} + \text{variance} + \text{irreducible error}$$

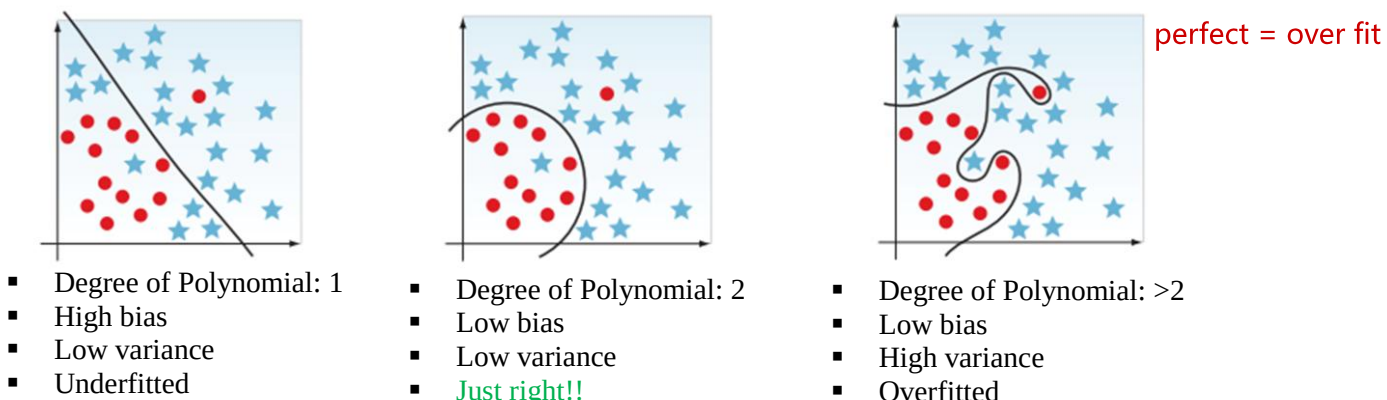
A model's generalization error can be expressed as the sum of three components: bias, variance and irreducible error.

- Bias** measures how far the model's average prediction is from the true target value.
 - Such models pay little attention to the training data and tend to oversimplify the problem.
 - High bias usually leads to underfitting, producing high training error and high test error.
 - Variance** measures how much the model's predictions would change if it were trained on a different dataset.
 - High-variance models are overly sensitive to small changes in the training data.
 - They tend to fit the training data very closely and fail to generalize well.
 - High variance leads to overfitting producing low training error but high test error.
 - Irreducible error** arises from noise inherent in the data itself, such as measurement errors or unobserved variables.
 - It cannot be eliminated by any model.
 - It can only be reduced by improving data quality, such as fixing faulty sensors, removing outliers, improving data collection procedures, etc.
- There is a fundamental tradeoff between bias and variance:
 - Increasing model complexity generally reduces bias but increases variance.
 - Decreasing model complexity generally increases bias but reduces variance.
 - The goal is to find a model that achieves the best balance, minimizing total generalization error.
 - Underfitting is characterized by high bias and low variance.
 - Overfitting is characterized by low bias and high variance.
 - Both overfitting and underfitting can occur in linear regression, logistic regression, and neural networks

Example 1: Recalling linear regression - predicting y from a single feature x .



Example 2: Recalling logistic regression - predicting y from a single feature x_1 and x_2 .



Addressing Overfitting

- Some of the common methods used to address overfitting are:
 - Increase Data
 - Early Stopping
 - Regularization

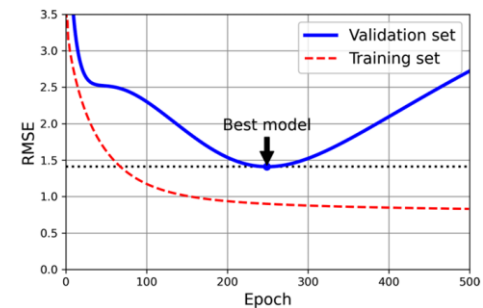
Increase Data

- Increasing the size and diversity of the training dataset helps the model learn more general patterns rather than memorizing noise. With more representative data, the model is less likely to overfit and typically achieves better generalization on unseen examples.
- However, acquiring more data is not always feasible due to domain constraints, data collection difficulties, privacy concerns, high costs, or limited availability of labeled samples.

Early Stopping

so that model does not learn validation data, so stop no matter the train data error

- In this technique, stop training as soon as the validation error reaches a minimum.
- As the epochs go by, the algorithm learns and its prediction error (RMSE) on the training set naturally goes down, and so does its prediction error on the validation set.
- However after a while the validation error stops decreasing and actually starts to go back up.
- This indicates that the model has started to overfit the training data.
- It is such a simple and efficient regularization technique that Geoffrey Hinton called it a “beautiful free lunch.”
- With Stochastic and Mini-batch Gradient Descent, the curves are not so smooth, and it may be hard to know whether you have reached the minimum or not. One solution is to stop only after the validation error has been above the minimum for some time (when you are confident that the model will not do any better), then roll back the model parameters to the point where the validation error was at a minimum.



Regularization

- A good way to reduce overfitting is to regularize the model; the fewer degrees of freedom it has, the harder it will be for it to overfit the data.
- For example, a simple way to regularize a polynomial model is to reduce the number of polynomial degrees.

Regularization by reducing the number of features

- Manually select which features to keep or use a feature selection algorithm.
- It might remove a less significant but unique feature, thus making the model a little weaker.

Regularization by constraining the weights

- If the hypothesis function is overfitting, the weight that some of the terms in the function carry can be reduced by increasing their cost.
- It is typically achieved by constraining the weights of the model.
- Regularization artificially amplifies bias error by penalizing an increase in a model's complexity.
- The idea is to keep all the features, but reduce the magnitude of parameters θ_j .
- Regularization works well when there are a lot of slightly useful features.
- Regularization can be done by modifying the cost function as follows:
 $J(\theta) = \text{error function} + \lambda \times \text{regularization function}$, where λ , or lambda, is the regularization parameter.
- This forces the learning algorithm to not only fit the data but also keep the model weights as small as possible.
- When number of features is very large it is really difficult to decide which parameters to regularize, so in practice all parameters, except θ_0 are regularized.
- Regularizing few or all θ parameters can smoothen out the output of the hypothesis function to reduce overfitting.
- The λ is a hyperparameter which controls how much you want to regularize the model.
- It can take value from 0 to any positive integer.
 - If $\lambda = 0$ then Regression is just normal Linear or Logistic Regression.
 - If λ is very large, it may smooth out the function too much and cause underfitting. All weights may end up very close to zero and the result is a flat line going through the data's mean.

- Hence, choice of this hyperparameter is a critical decision.
- This type of regularization has three types: Ridge Regularization, Lasso Regularization and Elastic Net.

Ridge Regularization

- In ridge regression, the cost function is altered by adding a penalty equivalent to square of the magnitude of the coefficients.
- Also called L2 regularization or Tikhonov regularization.
- Uses a regularization function/term equal to $\frac{\lambda}{n} \sum_{j=1}^m \theta_j^2$.
- Hence the cost function for ridge regularization is:

$$J(\theta) = \text{cost function} + \frac{\lambda}{n} \sum_{j=1}^m \theta_j^2$$

- Shrinks coefficients toward zero, but does not make them exactly zero
- Note that the regularization term should only be added to the cost function during training only. Once the model is trained, you want to evaluate the model's performance using the unregularized performance measure.
- It is important to scale the data before performing Ridge Regression, as it is sensitive to the scale of the input features. This is true of most regularized models.

Lasso Regularization

- Least Absolute Shrinkage and Selection Operator Regression (simply called Lasso Regression) uses absolute values of the parameters instead of the squared one.
- Also called L1 regularization.
- Uses a regularization function/term equal to $2\lambda \sum_{j=1}^m |\theta_j|$.
- Hence the cost function for lasso regularization is:

$$J(\theta) = \text{cost function} + 2\lambda \sum_{j=1}^m |\theta_j|$$

- Can shrink some coefficients to zero, effectively performing feature selection
- An important characteristic of Lasso Regression is that it tends to completely eliminate the weights of the least important features (i.e., set them to zero), i.e. the weights may be reduced to zero here.
- In other words, Lasso Regression automatically performs feature selection and outputs a sparse model (i.e., with few nonzero feature weights).
- Hence, it is very useful when trying to compress a model; otherwise, as per empirical results, L2 is usually preferred over it.

Elastic Net

- Elastic Net is a middle ground between Ridge Regression and Lasso Regression.
- The regularization term is a weighted sum of both Ridge and Lasso's regularization terms, and you can control the mix ratio r .
- The function equation is as follows:

$$J(\theta) = \text{cost function} + r \left(2\lambda \sum_{j=1}^m |\theta_j| \right) + (1-r) \left(\frac{\lambda}{n} \sum_{j=1}^m \theta_j^2 \right)$$

- When $r = 0$, Elastic Net is equivalent to Ridge Regression, and when $r = 1$, it is equivalent to Lasso Regression.

Which regularization technique to use?

- It is almost always preferable to have at least a little bit of regularization, so generally you should avoid plain Linear/Logistic Regression.
- Ridge is a good default, but if you suspect that only a few features are actually useful, you should prefer Lasso or Elastic Net since they tend to reduce the useless features' weights down to zero as we have discussed.
- In general, Elastic Net is preferred over Lasso since Lasso may behave erratically when the number of features is greater than the number of training instances or when several features are strongly correlated.